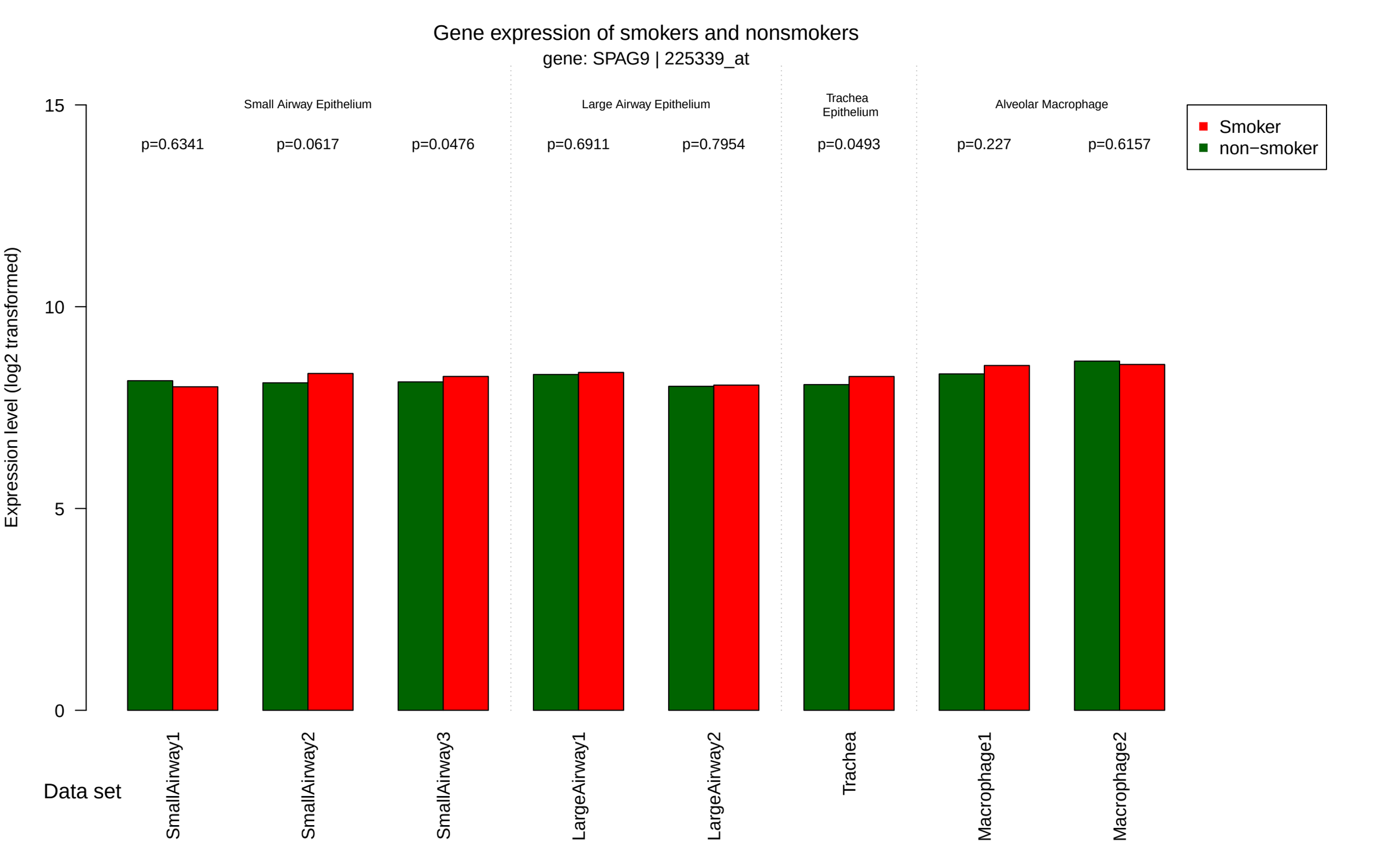
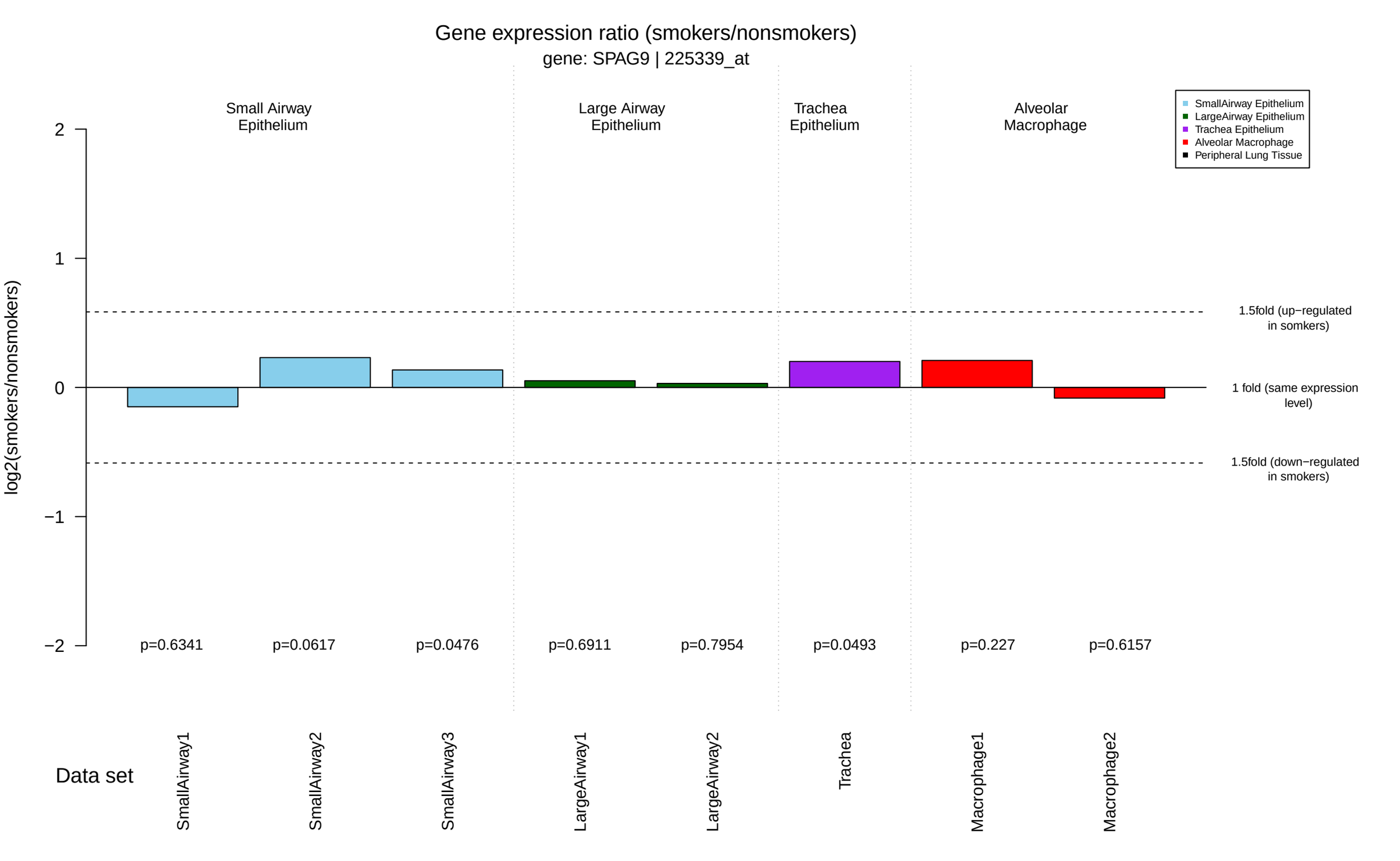
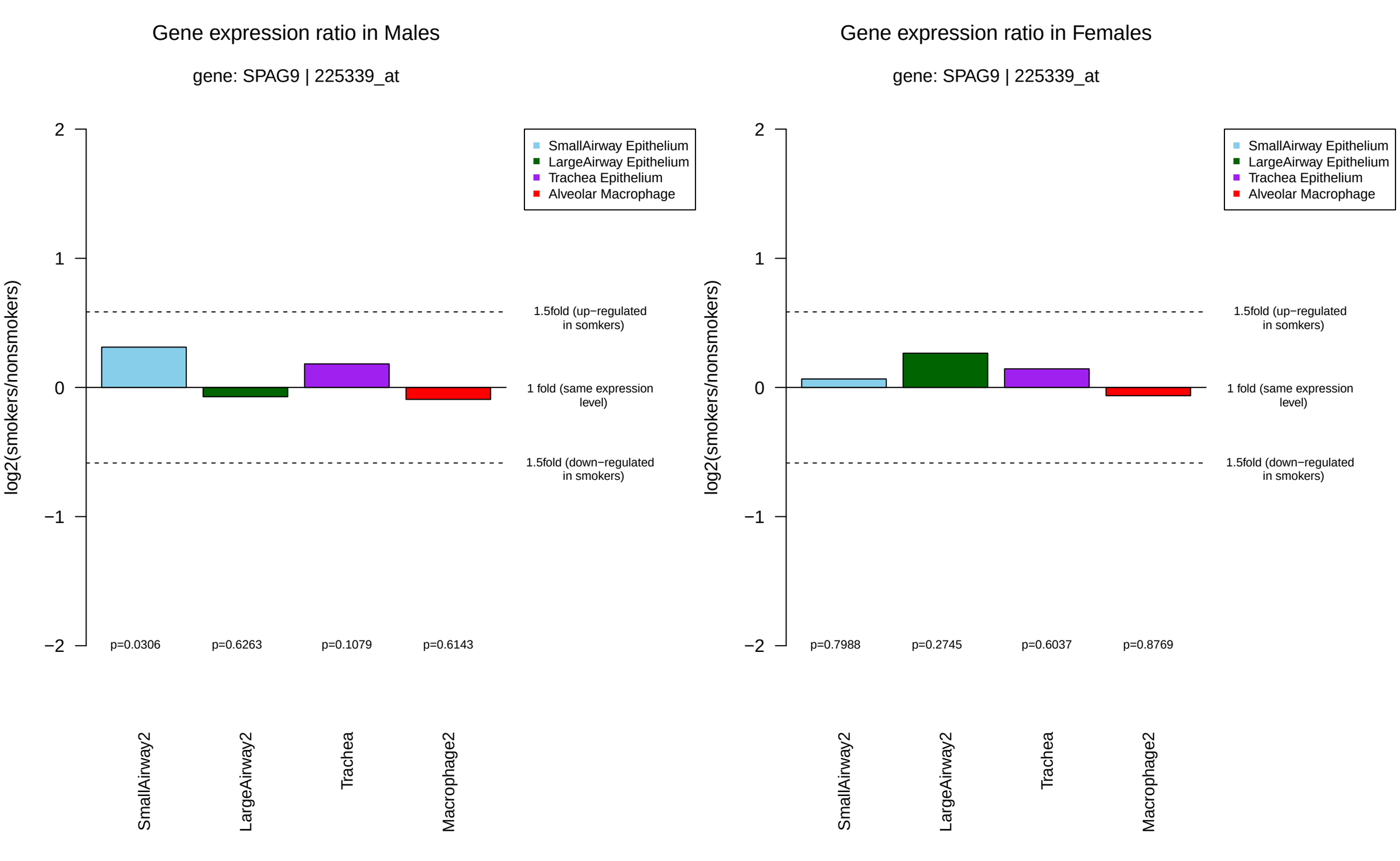
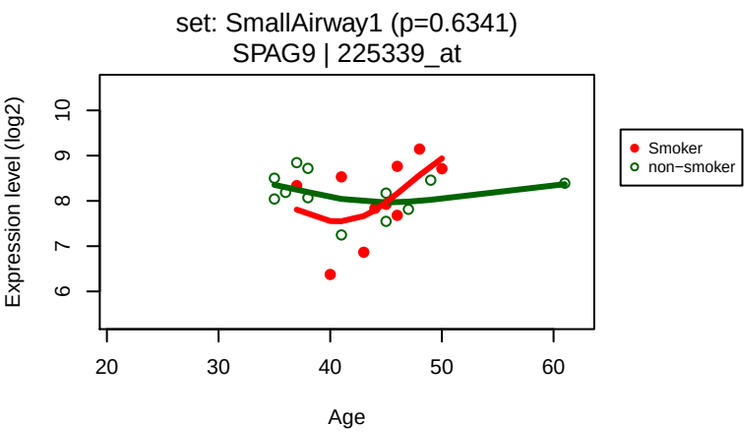
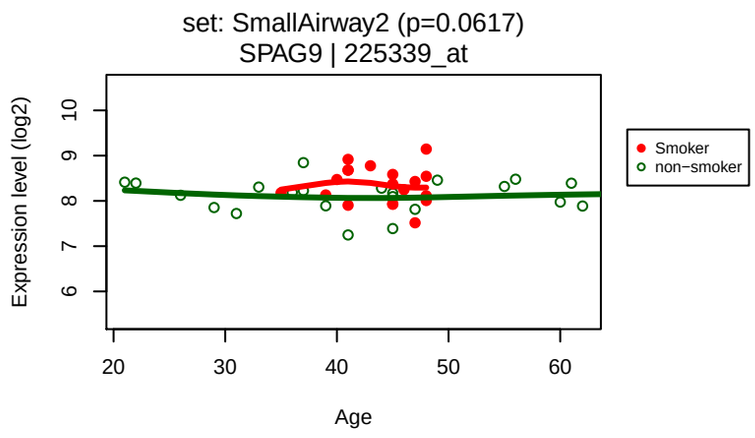
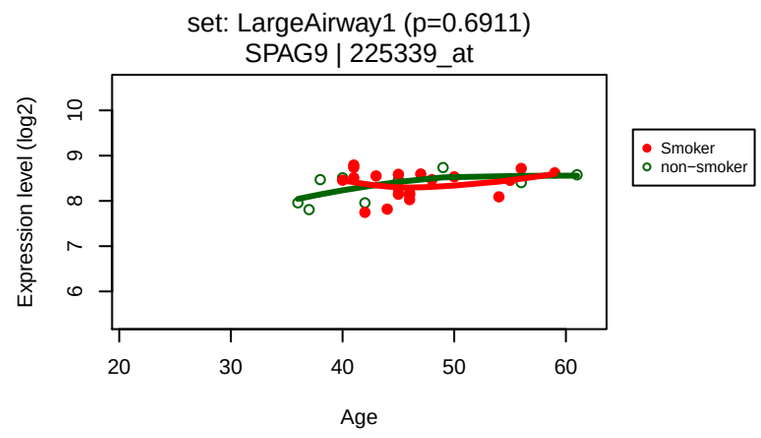

Gene expression of smokers and nonsmokers
gene: SPAG9 | 225339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6341
p=0.0617
p=0.0476
p=0.6911
p=0.7954
p=0.0493
p=0.227
p=0.6157
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

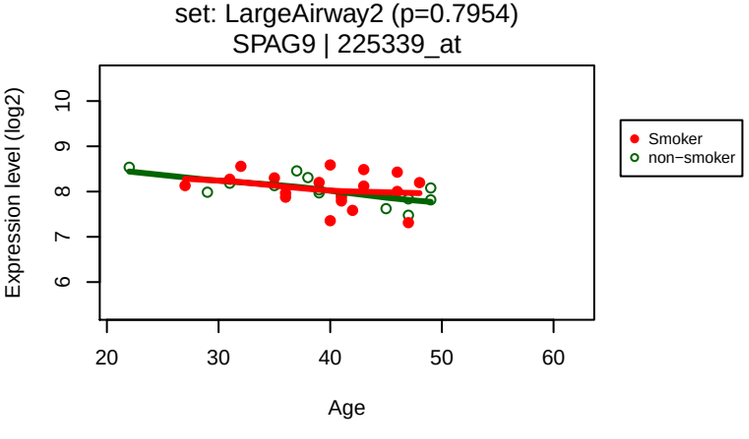

Gene expression ratio (smokers/nonsmokers)
gene: SPAG9 | 225339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6341
p=0.0617
p=0.0476
p=0.6911
p=0.7954
p=0.0493
p=0.227
p=0.6157
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

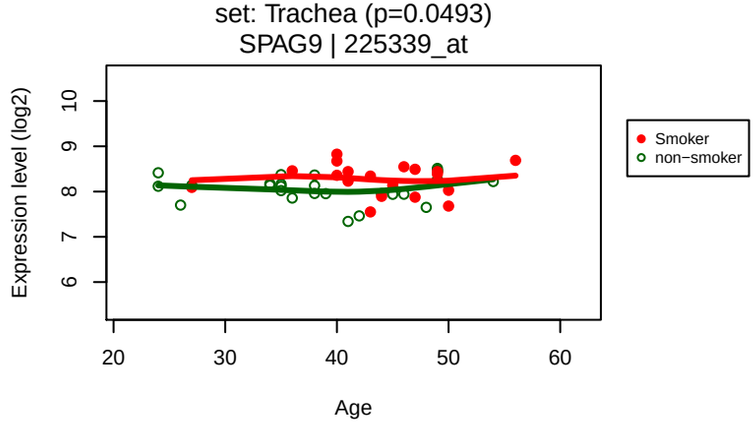

Gene expression ratio in Males
gene: SPAG9 | 225339_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0306
p=0.6263
p=0.1079
p=0.6143
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG9 | 225339_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7988
p=0.2745
p=0.6037
p=0.8769
SmallAirway2
LargeAirway2
Trachea
Macrophage2

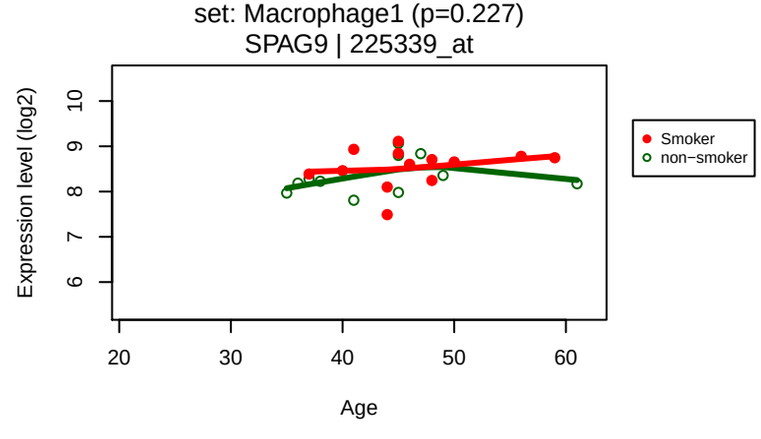

set: SmallAirway1 (p=0.6341)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker

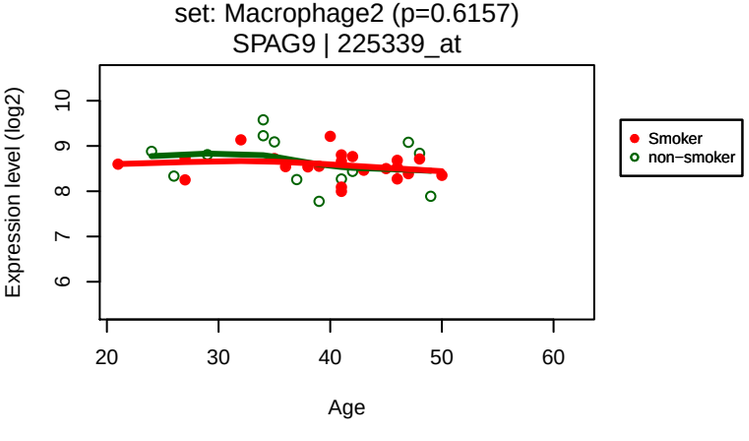

set: SmallAirway2 (p=0.0617)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker

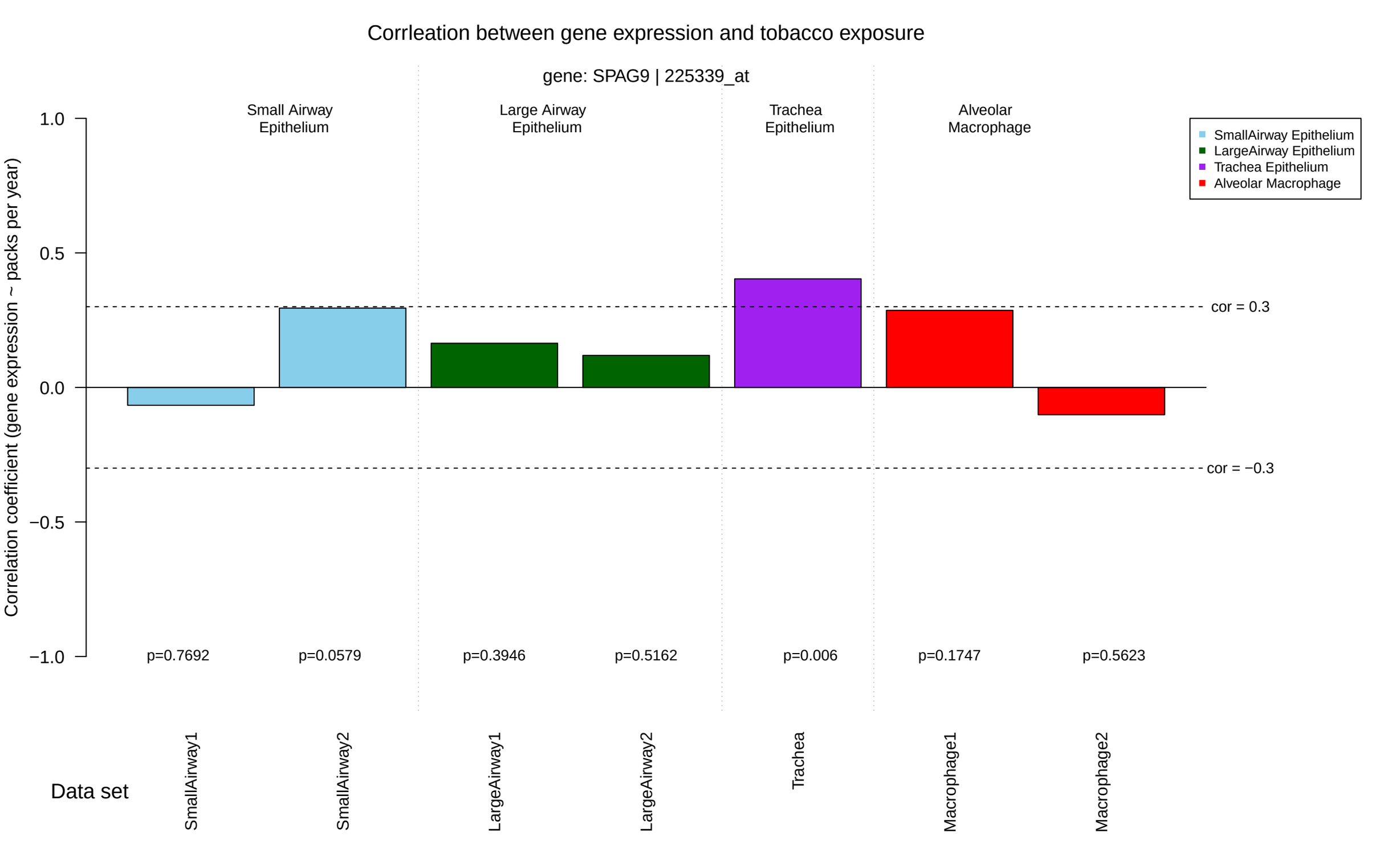

set: LargeAirway1 (p=0.6911)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7954)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0493)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.227)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6157)
SPAG9 | 225339_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG9 | 225339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7692
p=0.0579
p=0.3946
p=0.5162
p=0.006
p=0.1747
p=0.5623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2